Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Engineering & Technology

B.Tech. | Sem-5 | Winter-2025

Course Code	:	2301CS501	**Date**	:	08-10-2025
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** What is Computer network? Explain different types of computer networks with examples. **4**

(B) Define delay, loss, and throughput in computer networks. **3**

OR

Define Transmission media. Briefly explain guided and unguided transmission media.

(C) Draw and explain OSI reference model. **7**

OR

Explain different types of network topologies with neat diagrams.

Q.2 **(A)** Explain importance of application layer protocols. **4**

(B) Explain DHCP lease process in short. **3**

OR

Write short note on SNMP components.

(C) Explain HTTP request–response model with example. **7**

OR

Explain SMTP in detail with diagram.

Q.3 **(A)** A sender wants to transmit the following 16-bit words using checksum method: **4**
Word 1: 1010101010101010
Word 2: 1100110011001100
Find the checksum and show how the receiver verifies the data.

(B) Differentiate between UDP and TCP. **3**

OR

List and explain the services provided by the transport layer.

(C) Explain the principles of reliable data transfer protocols. **7**

OR

What is congestion? Explain congestion control mechanisms in TCP.

Q.4	**(A)**	Draw and explain the architecture of a router.	**4**
	(B)	What is subnetting? Explain with example.	**3**

OR

Compare addressing schemes of IPv4 and IPv6.

	(C)	Explain distance vector routing algorithm with example.	**7**

OR

Explain link state routing algorithm with steps.

Q.5	**(A)**	Write short note on polling and token passing.	**4**
	(B)	Differentiate between error detection and error correction.	**3**

OR

Differentiate between hub, switch, and bridge.

	(C)	Explain ALOHA and Slotted ALOHA.	**7**

OR

Explain CRC (Cyclic Redundancy Check) with an example.

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